This is an open access article published under an ACS AuthorChoice License, which permits copying and redistribution of the article or any adaptations for non-commercial purposes.

ACS Catalysis

pubs.acs.org/acscatalysis

Research Article

Enantioselective Cyanoborylation of Allenes by *N*-Heterocyclic Carbene-Copper Catalysts

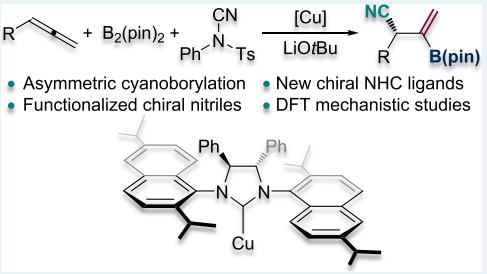
Zhenghua Li, Liang Zhang,* Masayoshi Nishiura, Gen Luo,* Yi Luo, and Zhaomin Hou*

Cite This: *ACS Catal.* 2020, 10, 11685–11692

Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information

ABSTRACT: The simultaneous incorporation of both a cyano group and a boryl unit into the C═C double bonds of allenes in a regio- and stereoselective fashion is of much interest and importance but remains a significant challenge. We report herein a copper-catalyzed chemo-, regio-, and enantioselective cyanoborylation of allenes, which afforded a family of valuable enantiopure *β*-boryl allyl nitriles. The high enantioselectivity was achieved by installing of appropriate substituents at the C2 and C6 positions of the naphthyl groups in our newly synthesized *N*-heterocyclic carbene (NHC) ligands. The reaction mechanism has been clarified by some stoichiometric reactions and computational studies. This work provides an inspiring example of the development of selective catalytic reactions for the synthesis of functional molecules through fine-tuning the ligands in catalysis.



KEYWORDS: *allenes, asymmetric catalysis, carbene ligand, copper, cyanoborylation*

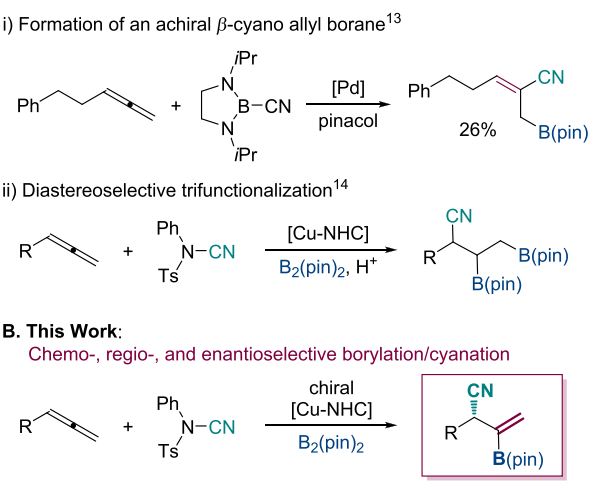
Downloaded via FUDAN UNIV on February 20, 2021 at 02:49:10 (UTC).
See https://pubs.acs.org/sharingguidelines for options on how to legitimately share published articles.

1. INTRODUCTION

The catalytic multifunctionalization of unsaturated C−C bonds is a straightforward and efficient strategy for the assembly of densely functionalized chemicals from simple starting materials. In view of the tremendous applications of organoboron compounds in various chemical transformations[1] and nitriles as versatile intermediates for the synthesis of many biologically active molecules, fine chemicals, dyes, pigments, and polyamides,[2,3] the direct cyanoborylation of allenes[4] in an enantioselective fashion, which would lead to the simultaneous installation of both a cyano group and a boryl group into a C═C double bond of the allene substrate, is of particular interest and importance. However, the difunctionalization of the 1,2-orthogonal C═C double bonds in allenes often suffers from regioselectivity problems. Besides, chemoselectivity issues can also arise due to the simultaneous existence of a number of reactive species. Although several copper-catalyzed enantioselective borylative functionalization reactions of allenes were reported recently,[5−11] the enantioselective cyanoborylation[12] of allenes has not been achieved yet.

In 2009, Suginome et al. disclosed an early example of palladium-catalyzed cyanoborylation of an allene compound via a metal-cyanide intermediate to give an achiral *β*-cyano allylborane in a relatively low yield (Scheme 1A, i).[13] In 2016, Montgomery et al. reported an elegant trifunctionalization of allenes for the synthesis of *β*,*γ*-diboryl nitriles through sequential cyanoborylation and hydroboration (Scheme 1A, ii).[14] In this transformation, suppressing the overaddition of a copper boryl species to the remaining C═C double bond in the initial cyanoborylation products was difficult. Therefore, the chemo-, regio-, and enantioselective cyanoborylation of allenes still remains a highly attractive research subject for the synthesis of valuable molecules containing multiple synthetically important functional groups, including a chiral nitrile unit, a boryl group, and a C═C double bond.

Scheme 1. Catalytic Cyanoborylation of Allenes



Received: July 10, 2020
Revised: August 31, 2020
Published: September 11, 2020



ACS Publications

© 2020 American Chemical Society

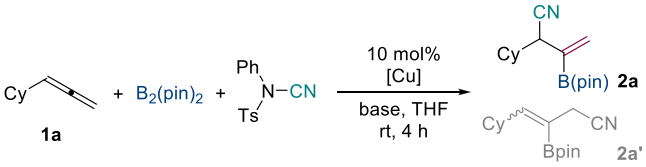
https://dx.doi.org/10.1021/acscatal.0c03018
ACS Catal. 2020, 10, 11685−11692

During our previous studies on the catalytic addition of copper boryl species to alkynes,[15] aldehydes,[16] and imines,[17] we became interested in the development of selective cyanoborylation of allenes through copper boryl catalysis. Herein, we report a copper-catalyzed chemo-, regio-, and enantioselective cyanoborylation of allenes, which afforded β-boryl allyl nitriles efficiently (Scheme 1B). The competing hydroboration of the remaining C═C double bonds after cyanoborylation was successfully suppressed. The systematic adjustment of the substituents on the C2 and C6 positions of the naphthalene moieties in our newly prepared chiral *N*-heterocyclic carbene (NHC) ligands enabled the direct synthesis of valuable enantiopure β-boryl allyl nitriles, which could be readily elaborated into useful organic molecules. This method shows excellent functional group tolerability and is easy to scale up. Computational studies revealed that the regioselectivity of borocupration is governed by both the steric hindrance and electronic effects. The enantioselectivity of cyanation is attributed to the steric effects of the substituted naphthyl wingtips of the NHC ligand. The current reaction represents an inspiring example of fine-tuning of a ligand enabling selective catalysis for the efficient synthesis of complex molecules from simple feedstocks.

2. RESULTS AND DISCUSSION

We began the research study with the investigation of various ligands for the copper-catalyzed cyanoborylation of cyclohexylallene **1a** with bis(pinacolato)diboron ($B_2(pin)_2$)[18] and electrophilic cyanation agent *N*-cyano-*N*-phenyl-*p*-toluenesulfonamide (NCTS)[19] (Table 1). Among all the tested copper catalysts, the NHC-copper complex [(SIMes)CuCl] (SIMes = 1,3-bis(2,4,6-trimethylphenyl)imidazolin-2-ylidene) resulted in the formation of desired cyanoborylation product **2a** in high yield and regioselectivity (entry 3). The formation of only 4% of regioisomer **2a′** was detected. Remarkably, the catalyst loading could be reduced to 5 mol % without significant erosion of the yield (entry 4). It is noteworthy that the further borylation of the C═C double bond in **2a** was successfully suppressed (less than 4% of the trifunctionalization product was detected by ^{1}H NMR analysis), in contrast with what was observed previously in the trifunctionalization of allenes.[14] Both [(IPr)CuCl] (IPr = 1,3-bis(2,6-diisopropylphenyl)imidazol-2-ylidene) and [(IMes)CuCl] (IMes = 1,3-bis(2,4,6-trimethylphenyl)imidazol-2-ylidene) gave lower yields, albeit comparable regioselectivities were observed (entries 1 and 2). LiO*t*Bu is the best among the tested bases (entries 4−6). The combinations of CuCl with a series of phosphine ligands (entries 7−14) were found to be inferior to [(SIMes)CuCl].

Table 1. Screening of Reaction Conditions for the Cyanoborylation of Allene 1a[a]



entry	[Cu]	base	yield (%)[b]	2a/2a′[e]
1	[(IPr)CuCl]	LiO*t*Bu	9	97:3
2	[(IMes)CuCl]	LiO*t*Bu	63	96:4
3	[(SIMes)CuCl]	LiO*t*Bu	96(79)[c]	96:4
4	[(SIMes)CuCl][d]	LiO*t*Bu	94	96:4
5	[(SIMes)CuCl][d]	NaO*t*Bu	77	97:3
6	[(SIMes)CuCl][d]	KO*t*Bu	53	98:2
7	CuCl/PCy_3	LiO*t*Bu	30	72:28
8	CuCl/2PCy_3	LiO*t*Bu	66	82:18
9	CuCl/4PCy_3	LiO*t*Bu	56	83:17
10	CuCl/2*t*Bu_2MeP	LiO*t*Bu	25	72:28
11	CuCl/dcpm	LiO*t*Bu	6	74:26
12	CuCl/dcpe	LiO*t*Bu	65	98:2
13	CuCl/BINAP	LiO*t*Bu	7	96:4
14	CuCl/*t*Bu-Xantphos	LiO*t*Bu	trace	

[a]Reaction conditions: [Cu] (10 mol %), base (1.4 equiv), THF (2.0 mL), rt, and 15 min. Then, **1a** (0.5 mmol), $B_2(pin)_2$ (1.2 equiv), and NCTS (1.2 equiv) in THF (8.0 mL) were added, rt, 4 h. [b]NMR yields. [c]Isolated yield. [d][Cu] (5 mol %) was used. [e]Determined by GC−MS. dcpm = Bis(dicyclohexylphosphino)methane. dcpe = Bis(dicyclohexylphosphino)ethane. *t*Bu-Xantphos = 9,9-dimethyl-4,5-bis(di-*tert*-butylphosphino)xanthene.

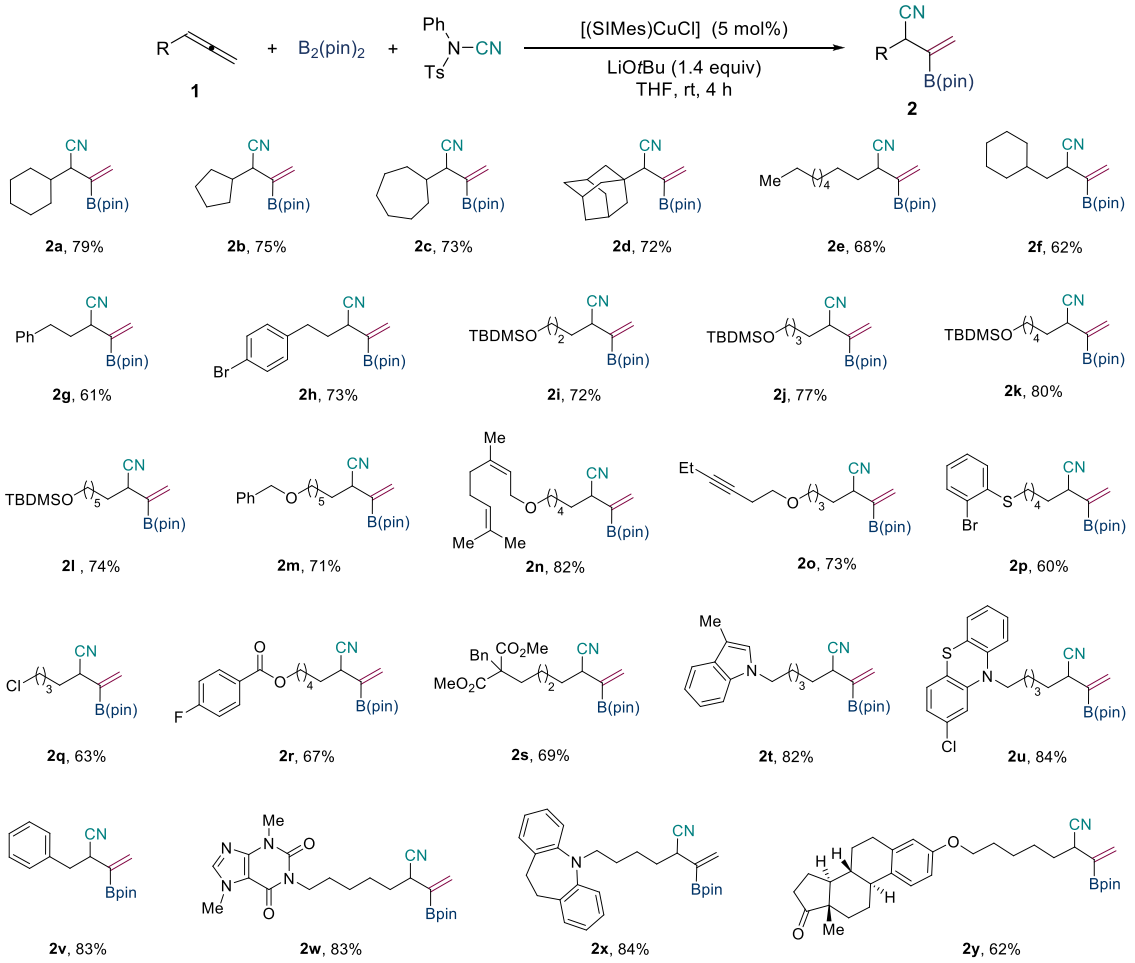
Under the optimized reaction conditions (Table 1, entry 4), we investigated the substrate scope of the [(SIMes)CuCl]-catalyzed cyanoborylation of a wide range of allenes (Table 2). The reactions of a variety of terminal allenes afforded the desired cyanoborylation products **2a−y** in good to excellent yields selectively. Remarkably, functional groups such as silyl ether, benzyl ether, thioether, alkene, alkyne, ester, heterocycle, and halides (Br, Cl, and F) were compatible with the current reaction conditions.

Considering the great importance of chiral nitriles,[2,3c,3d] we then investigated a large series of C2 symmetric NHC ligands based on a chiral diphenylethylene diamine backbone for the enantioselective cyanoborylation of allene **1a** (Table 3). In the presence of 5 mol % CuCl, the reactions using *N*-phenyl-substituted NHC ligands **L1−6** generally afforded chiral nitrile product (*S*)-**2a** regioselectively in good yields (Table 3, entries 1−6). It is also encouraging that a moderate enantiomeric ratio was achieved in the case of **L6** (75:25 er, entry 6). When employing NHC ligand **L7** with naphthyl wingtips,[20] a slight improvement of enantioselectivity was observed (77:23 er, entry 7). It was proven that the introduction of methyl (**L8**) or isopropyl (**L9**) to the C2 position of naphthyl groups led to the same enantiomeric outcome (entries 8 and 9). In order to inspect the substituent effect at other positions, we synthesized new NHC ligands **L10−12**, which features an alkyl group on the C6 position of a naphthyl moiety. The reaction using **L10** only showed a subtle improvement in the enantioselectivity (79:21 er, entry 10). Replacing the methyl groups in **L10** with isopropyl groups, **L11** led to a marked increase in the enantiomeric ratio (88:12 er, entry 11). The introduction of bulkier adamantyl to the C2 position of naphthyl groups, ligand **L12** resulted in a slightly decreased enantiomeric ratio (86:14 er, entry 12). The effect of reaction temperature was then examined by using ligand **L11** (entries 13−16). Performing the reaction at −5 °C resulted in a higher enantiomeric ratio (93:7 er, entry 14). The further examination of copper sources indicated that the best enantioselectivity (96:4 er) could be obtained when $[Cu(CH_3CN)_4]PF_6$ was used (entries 17−19). While 10 mol % catalyst loading furnished (*S*)-**2a** in high yield, a lower yield was observed when 5 mol % catalyst was used (entry 20).

After identifying **L11** as the best candidate of ligands, we then investigated the substrate scope of the asymmetric cyanoborylation of allenes under the optimized reaction conditions (Table 3, entry 19). The results are summarized in Table 4. Using the substrates previously described in Table 2, the new enantiomerically selective catalyst system gave good to excellent

https://dx.doi.org/10.1021/acscatal.0c03018
ACS Catal. 2020, 10, 11685−11692

Table 2. Substrate Scope of [(SIMes)CuCl]-Catalyzed Cyanoborylation of Allenes[a]



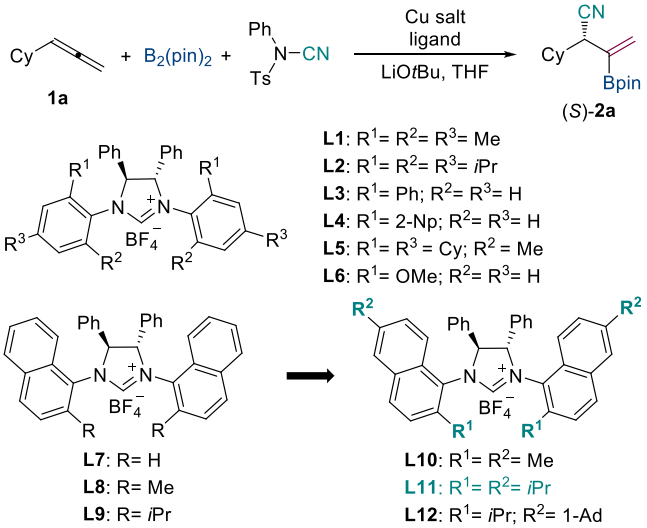
[a]Reaction conditions: [(SIMes)CuCl] (5 mol %), **1** (0.5 mmol), $B_2(pin)_2$ (1.2 equiv), NCTS (1.2 equiv), LiO*t*Bu (1.4 equiv), THF (10.0 mL), rt, and 4 h. Isolated yields are shown. TBDMS = *tert*-butyldimethylsilyl.

yields with relatively high enantioselectivity (89−96% yield of the primary enantiomer). To demonstrate the practicality of this novel method for the synthesis of chiral nitriles, a 2.0 mmol scale reaction using only 5 mol % catalyst was carried out, affording (*S*)-**2a** in 78% yield and 96:4 er.

To demonstrate their synthetic utility, we explored a series of reactions of the obtained densely functionalized chiral nitriles, affording a number of valuable molecules (Scheme 2). First, copper-mediated brominations of (*S*)-**2a** or (*R*)-**2d** were found to yield vinyl bromide containing chiral nitriles (*R*)-**3a** or (*R*)-**3b** with the enantioselectivity retained (Scheme 2A).[21] The absolute configuration of (*R*)-**3b** was determined by X-ray crystallographic analysis.[22] The vinyl boronic ester group in (*S*)-**2a** could also react with allyl alcohol to afford chiral nitrile (*S*)-**4** with a 93:7 enantiomeric ratio (Scheme 2B).[23] Palladium-catalyzed cross-coupling reaction of (*S*)-**2a** with methyl 4-bromobenzoate resulted in the formation of tetra-substituted alkene **5** with the isomerization of a C═C double bond was observed (Scheme 2C).[24]

To gain some insights into the reaction mechanism of the present catalytic process, we then examined the stoichiometric reaction of a NHC-copper boryl complex [(IPr)CuB(pin)], which was obtained from the reaction of [(IPr)Cu(O*t*Bu)] with $B_2(pin)_2$,[25] with 1.0 equiv of cyclohexylallene **1a** (Scheme 3). Allyl cuprate **6**[26] through the borocupration of the unsubstituted

Table 3. Screening of Reaction Conditions for the Asymmetric Cyanoborylation of Allene 1a[a]



entry	Cu salt	L	*T* (°C)	*t* (h)	yield (%)[b]	er[c]
1	CuCl	**L1**	rt	4	83	74:26
2	CuCl	**L2**	rt	4	92	57:43
3	CuCl	**L3**	rt	4	85	67:33
4	CuCl	**L4**	rt	4	45	71:29

https://dx.doi.org/10.1021/acscatal.0c03018
ACS Catal. 2020, 10, 11685−11692

Table 3. continued

entry	Cu salt	L	T (°C)	t (h)	yield (%)[b]	er[c]
5	CuCl	**L5**	rt	4	90	74:26
6	CuCl	**L6**	rt	4	81	75:25
7	CuCl	**L7**	rt	4	51	77:23
8	CuCl	**L8**	rt	4	98	77:23
9	CuCl	**L9**	rt	4	96	77:23
10	CuCl	**L10**	rt	4	98	79:21
11	CuCl	**L11**	rt	14	75	88:12
12	CuCl	**L12**	rt	12	78	86:14
13	CuCl	**L11**	0	24	95	88:12
14[d]	CuCl	**L11**	−5	48	85	93:7
15[d]	CuCl	**L11**	−10	48	96	93:7
16[d]	CuCl	**L11**	−20	48	65	89:11
17[d]	CuI	**L11**	−5	48	69	93:7
18[d]	CuBr	**L11**	−5	48	52	89:11
19[d]	$[Cu(CH_3CN)_4]PF_6$	**L11**	−5	48	98(78)[e]	96:4
20	$[Cu(CH_3CN)_4]PF_6$	**L11**	−5	48	73	96:4

[a]Reaction conditions: **1a** (0.5 mmol), NCTS (1.2 equiv), $B_2(pin)_2$ (1.2 equiv), LiO*t*Bu (1.4 equiv), Cu salt/L (5 mol %), and THF (10.0 mL). [b]NMR yields. [c]The enantiomeric ratios were determined by HPLC analysis on a chiral stationary phase. [d]A catalyst (10 mol %) was used. [e]Isolated yield.

Scheme 2. Synthetic Applications of Chiral Nitriles

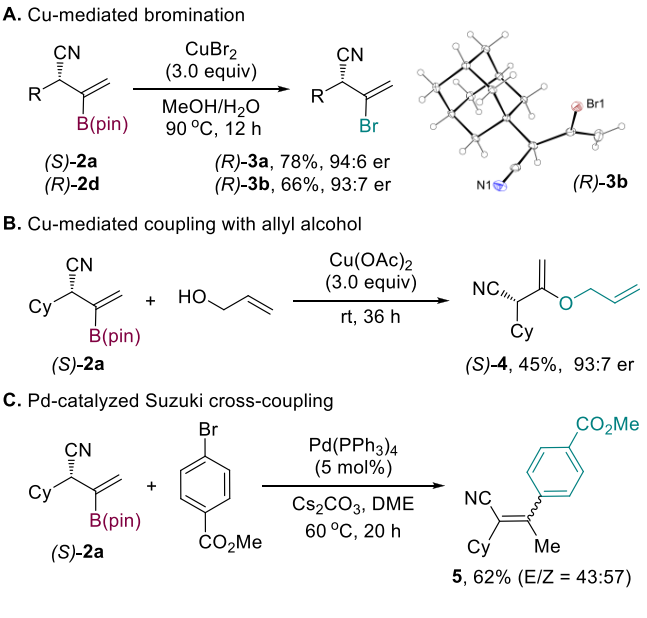
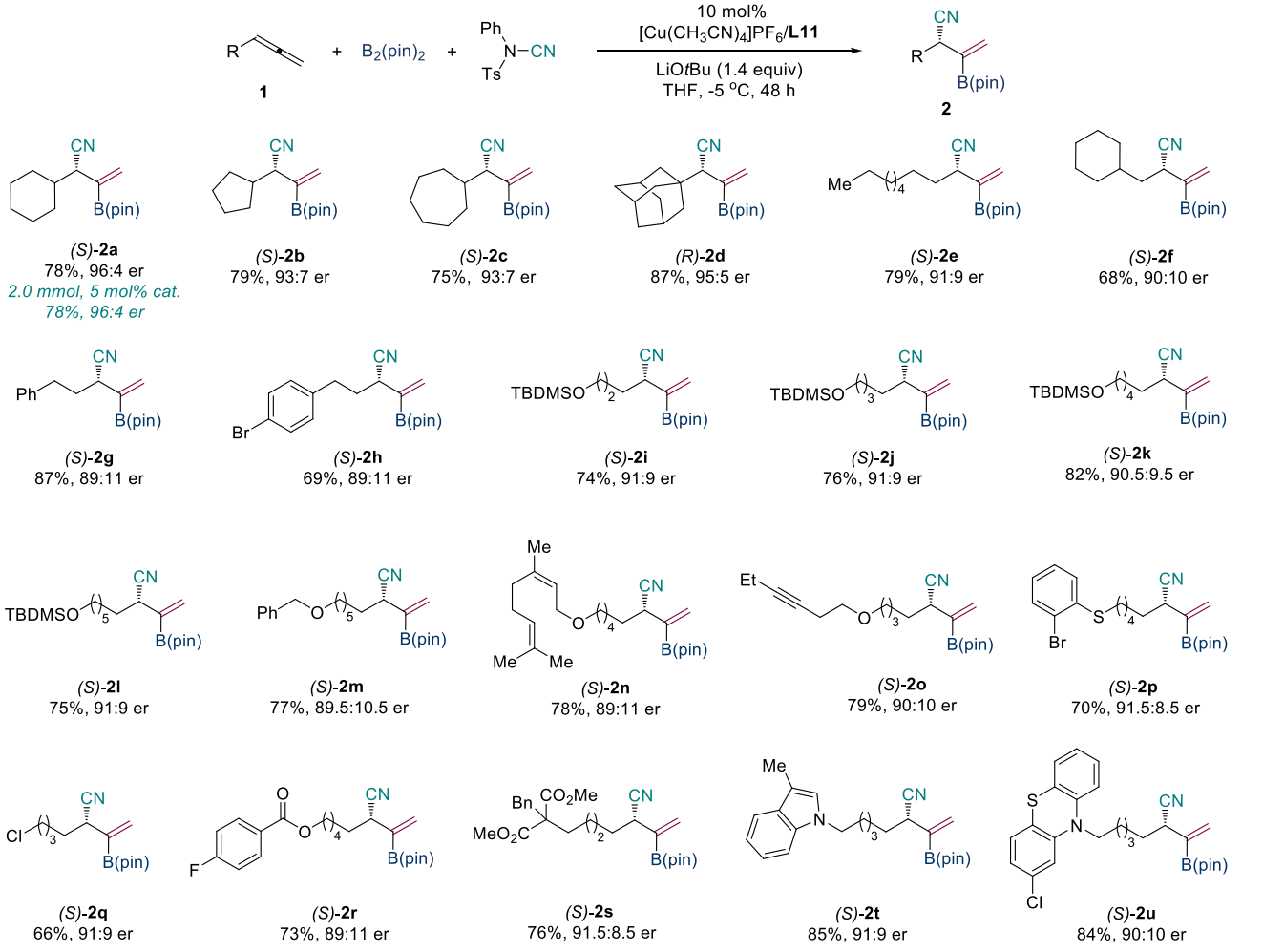


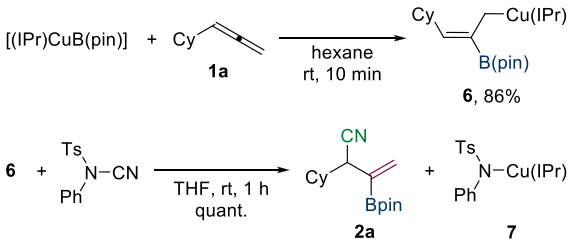
Table 4. Substrate Scope of Copper-Catalyzed Asymmetric Cyanoborylation of Allenes[a]



[a]Reaction conditions: $[Cu(CH_3CN)_4]PF_6$ (10 mol %), **L11** (10 mol %), allenes **1** (0.5 mmol), $B_2(pin)_2$ (1.2 equiv), NCTS (1.2 equiv), LiO*t*Bu (1.4 equiv), THF (10.0 mL), −5 °C, and 48 h. Isolated yields are shown. The enantiomeric ratio values were determined by HPLC analysis on a chiral stationary phase. TBDMS = *tert*-butyldimethylsilyl.

https://dx.doi.org/10.1021/acscatal.0c03018
ACS Catal. 2020, 10, 11685−11692

Scheme 3. Stoichiometric Reactions



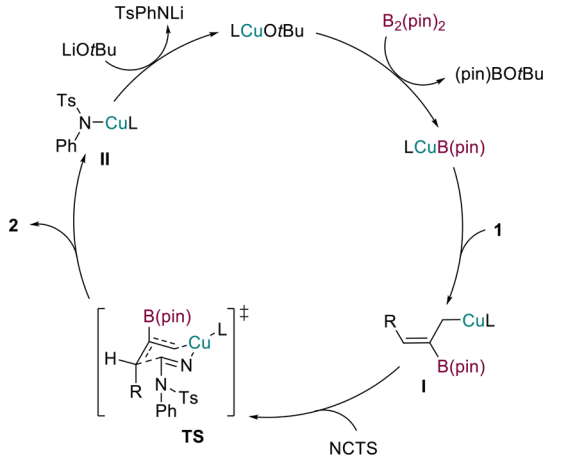


Figure 1. Proposed reaction mechanism for the copper-catalyzed cyanoborylation of allenes.

allene π-system was isolated in 86% yield. The reaction of **6** with 1 equiv of NCTS in THF directly afforded cyanoborylation product **2a** and copper imide **7** quantitatively.

A possible reaction mechanism for this NHC-copper catalyzed selective cyanoborylation of allenes is proposed as shown in Figure 1. The initial reaction between catalyst precursors and LiO*t*Bu would afford copper alkoxide [LCuO*t*Bu], which upon reaction with $B_2(pin)_2$ could generate copper boryl complex [LCuB(pin)]. Its subsequent regioselective borocupration of allene substrate **1** would furnish allyl cuprate **I**. Further attack of **I** to the C−N triple bond of NCTS with allylic transposition via transition state **TS** would release β-boryl allyl nitrile product **2** along with copper imide **II**. Subsequent transmetallation between **II** and LiO*t*Bu would regenerate [LCuO*t*Bu].

To get a deep understanding of the mechanistic details and the selectivities of the current NHC-copper catalyzed cyanoborylation reaction, DFT calculations were performed on the reaction of cyclohexylallene **1a** with $B_2(pin)_2$ and NCTS catalyzed by the SIMes-ligated copper catalyst. As shown in Figure 2, the coordination of an allene substrate **1a** to [(SIMes)CuB(pin)] (**A**) could give the π-complex, and then, the borocupration took place through the insertion of a C═C double bond of allene substrate **1a** into a Cu−B bond. During this step, there are four insertion manners, viz., 1,2-insertion (**TS1**), 2,1-insertion (**TS1**^{a}), 3,2-insertion (**TS1**^{b}), and 2,3-insertion (**TS1**^{c}),[27] which will afford the corresponding intermediates **C**, **C**^{a}, **C**^{b}, and **C**^{c}. The results suggested that the 1,2-insertion via **TS1** is the most favorable manner with the lowest energy barrier of 15.7 kcal/mol, which is in agreement with experimentally observed regioselectivity. This is mainly due to the smallest sterics between allene substrate **1a** and the SIMes ligand for the 1,2-insertion manner in comparison with other pathways (Figure S2). In addition, the nucleophilic boryl group is more likely to couple with the π-electron deficient sp-C over the sp^2-C of allene **1a**.[25c] Therefore, both the steric and electronic factors account for the regioselectivity of 1,2-insertion via **TS1**.[14b] Similarly, complex **C** is more thermodynamically stable than the other isomers.

Subsequently, preferred intermediate **C** would react with NCTS via a potential stereoselective fashion. Four insertion

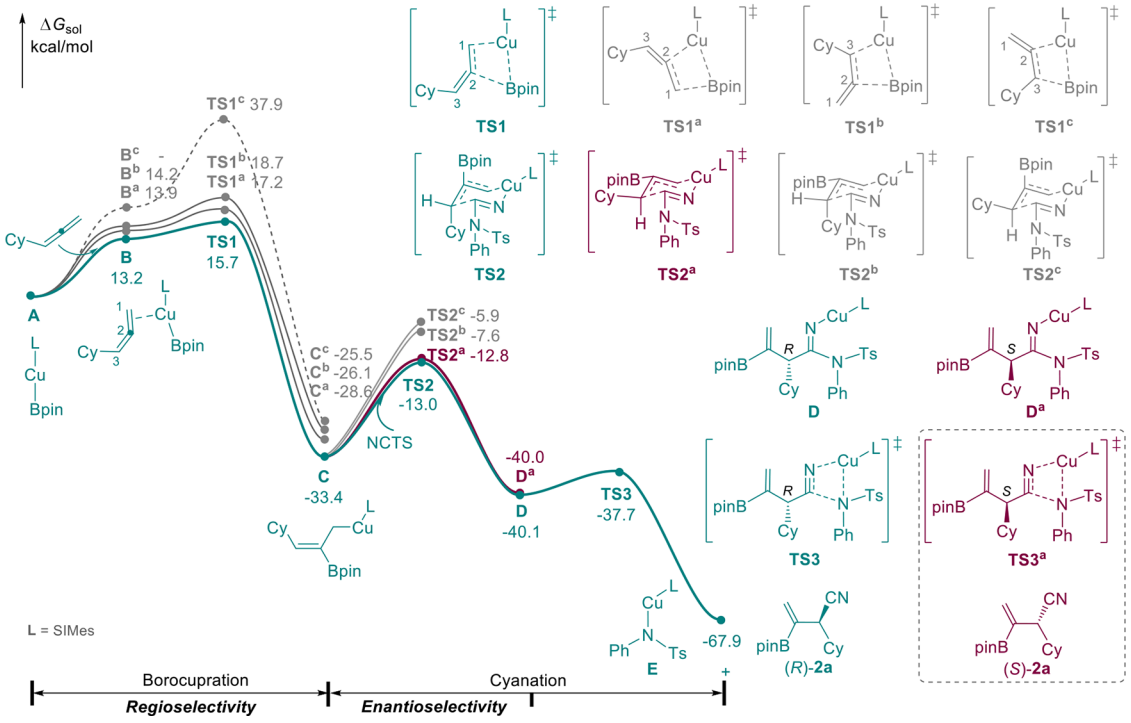


Figure 2. Computational analysis of regioselective cyanoborylation of allenes by the SIMes-Cu catalyst. It is calculated at the M06/6-311+G(d,p) &SDD (SMD, THF)//B3LYP/6-31G(d)&LanL2DZ level of theory. See the Supporting Information for more computational details.

https://dx.doi.org/10.1021/acscatal.0c03018
ACS Catal. 2020, 10, 11685−11692

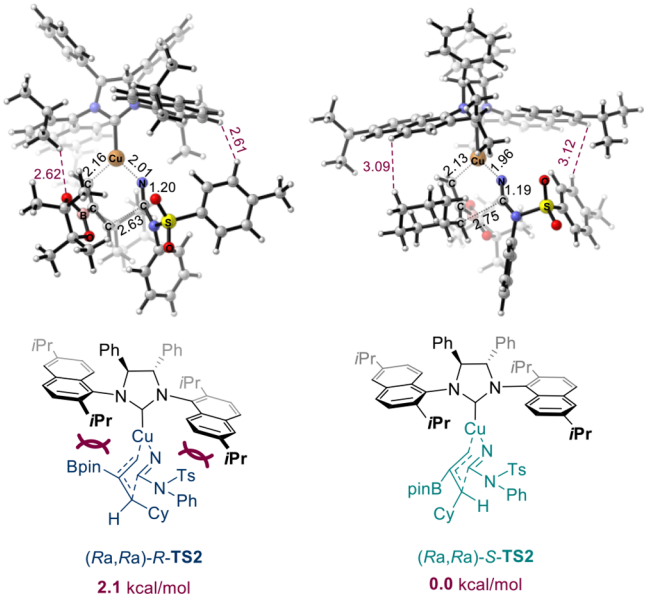


Figure 3. Calculated transition states of (*Ra*,*Ra*)-*R*-**TS2** and (*Ra*,*Ra*)-*S*-**TS2**. The relative free energies and interatomic distances are given in kcal/mol and Å, respectively.

manners of **C** into a C−N triple bond in NCTS were considered by six-member ring conformation. The computational results suggested that the two most favorable manners of **TS2** and **TS2**[a], in which the Cy and Bpin groups are located on the opposite side, are almost with the same energy barrier (20.4 vs 20.6 kcal/mol) and thus will give the racemic products (both (*R*)-**2a** and (*S*)-**2a**). This is consistent with the experimental observation when using the achiral SIMes ligand. The much higher energy barriers for the other two transition states **TS2**[b] and **TS2**[c] (25.8 and 27.5 kcal/mol, respectively) are mainly due to the significantly steric repulsion between the Cy and Bpin groups on the same side (Figure S3). Finally, favored imido copper intermediate **D** could easily transform to **E** and release product (*R*)-**2a** via transition state **TS3**. This step shows an energy barrier of 2.4 kcal/mol and is significantly exergonic by 27.8 kcal/mol. Similarly, intermediate **D**[a] could give product (*S*)-**2a**.[28]

To further understand the enantioselectivity in the case of chiral NHC ligand **L11**, we performed the calculations on the enantioselectivity-controlling step. It should be noted that there are three diastereomers of ligand precursor **L11**. Thus, three diastereomers of copper boryl species [L*CuB(pin)], i.e., (*Ra*,*Ra*)-**A**, (*Ra*,*Sa*)-**A**, and (*Sa*,*Sa*)-**A** were first calculated, and it was found that they have almost the same thermodynamic stability (Scheme S1).

The calculated results with the (*Ra*,*Ra*)-**L11** ligand show that the transition state (*Ra*,*Ra*)-*S*-**TS2** (leading to product (*S*)-**2a**) is more favorable than (*Ra*,*Ra*)-*R*-**TS2** (leading to (*R*)-**2a**) by 2.1 kcal/mol, which is well in line with the experimentally observed enantioselectivity (Figure 3). Through careful examination of the two cyanation transition states, the enantioselectivity was found to be mainly originated from the steric repulsion between the substituents of the substrates and the chiral NHC ligand. As shown in Figure 3, in (*Ra*,*Ra*)-*R*-**TS2**, the shortest interatomic distance between an NHC ligand and a boryl group or tosylate group is about 2.6 Å, while the similar shortest interatomic distance is more than 3.0 Å in the favored (*Ra*,*Ra*)-*S*-**TS2**. As for the cases of (*Ra*,*Sa*)-**L11** and (*Sa*,*Sa*)-**L11** ligands, the cyanations via an *S*-**TS2** manner are also more preferred than an *R*-**TS2** manner by 2.6 and 0.9 kcal/mol, respectively. Therefore, the same enantioselectivity could be obtained with three diastereomers of ligand **L11**.

3. CONCLUSIONS

In summary, we have developed a copper-catalyzed chemo-, regio-, and enantioselective cyanoborylation of allenes by the use of a new chiral NHC ligand, which highlights the isopropyl substituents at the C2 and C6 positions of the naphthyl wingtips. The reactions of a wide range of allenes afforded the enantiopure β-boryl allyl nitriles, which show broad synthetic potentials. The unwanted overaddition of copper boryl species to the remaining boron-activated C=C double bonds after cyanoborylation was successfully suppressed. Stoichiometric reactions of NHC-Cu complexes were also carried out to prove some mechanistic aspects. Computational studies found that the stability of the borocupration transition states governs the regioselectivity of the current reaction. Further DFT calculations also revealed that the enantioselectivity was controlled by the steric effects of chiral NHC ligand **L11**. This method sets up an enlightening model for the development of selective catalytic processes through fine-tuning ligand design.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.0c03018.

Experimental and computational details and characterization data for all new compounds (PDF)

X-ray crystallographic data of (*R*)-**3b** (CIF)

■ AUTHOR INFORMATION

Corresponding Authors

Liang Zhang − *Organometallic Chemistry Laboratory, RIKEN Cluster for Pioneering Research, Wako, Saitama 351-0198, Japan; Advanced Catalysis Research Group, RIKEN Center for Sustainable Resource Science, Wako, Saitama 351-0198, Japan;* orcid.org/0000-0002-5661-3557; Email: lzhang@riken.jp

Gen Luo − *Institute of Physical Science and Information Technology, Anhui University, Hefei 230601, China;* orcid.org/0000-0002-5297-6756; Email: luogen@ahu.edu.cn

Zhaomin Hou − *Organometallic Chemistry Laboratory, RIKEN Cluster for Pioneering Research, Wako, Saitama 351-0198, Japan; Advanced Catalysis Research Group, RIKEN Center for Sustainable Resource Science, Wako, Saitama 351-0198, Japan; State Key Laboratory of Fine Chemicals, School of Chemical Engineering, Dalian University of Technology, Dalian 116024, China;* orcid.org/0000-0003-2841-5120; Email: houz@riken.jp

Authors

Zhenghua Li − *Organometallic Chemistry Laboratory, RIKEN Cluster for Pioneering Research, Wako, Saitama 351-0198, Japan;* orcid.org/0000-0002-2291-4511

Masayoshi Nishiura − *Organometallic Chemistry Laboratory, RIKEN Cluster for Pioneering Research, Wako, Saitama 351-0198, Japan; Advanced Catalysis Research Group, RIKEN Center for Sustainable Resource Science, Wako, Saitama 351-0198, Japan;* orcid.org/0000-0003-2748-9814

https://dx.doi.org/10.1021/acscatal.0c03018
ACS Catal. 2020, 10, 11685−11692

Yi Luo − *State Key Laboratory of Fine Chemicals, School of Chemical Engineering, Dalian University of Technology, Dalian 116024, China;* orcid.org/0000-0001-6390-8639

Complete contact information is available at: https://pubs.acs.org/10.1021/acscatal.0c03018

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

Financial support by JSPS KAKENHI grant number JP17H06451 (Z.H.) (Hybrid Catalysis) is gratefully acknowledged. G.L. thanks the Innovation and Entrepreneurship Project of Overseas Returnees in Anhui Province (2020LCX006). The authors also thank RIKEN's HOKUSAI system, the Network and Information Center of Dalian University of Technology, and the High-Performance Computing Platform of Anhui University for computational resources.

■ REFERENCES

(1) For reviews, see (a) Miyaura, N.; Suzuki, A. Palladium-catalyzed cross-coupling reactions of organoboron compounds. *Chem. Rev.* **1995**, *95*, 2457−2483. (b) Hall, D. G. Ed. *Boronic Acids*; Wiley-VCH: Weinheim, 2005, 1−436.

(2) (a) Fleming, F. F.; Yao, L.; Ravikumar, P. C.; Funk, L.; Shook, B. C. Nitrile-containing pharmaceuticals: efficacious roles of the nitrile pharmacophore. *J. Med. Chem.* **2010**, *53*, 7902−7917. (b) Rappoport, Z. Ed. *The Chemistry of the Cyano Group*; Wiley: Weinheim, 1971, 1−946.

(3) For selected reviews, see (a) RajanBabu, T. V. B. Hydrocyanation of alkenes and alkynes. *Org. React.* **2011**, *75*, 1−72. (b) Tolman, C. A.; McKinney, R. J.; Seidel, W. C.; Druliner, J. D.; Stevens, W. R. Homogeneous nickel-catalyzed olefin hydrocyanation. *Adv. Catal.* **1985**, *33*, 1−46. (c) Khan, N. H.; Kureshy, R. I.; Abdi, S. H. R.; Agrawal, S.; Jasra, R. V. Metal catalyzed asymmetric cyanation reactions. *Coord. Chem. Rev.* **2008**, *252*, 593−623. (d) Kurono, N.; Ohkuma, T. Catalytic asymmetric cyanation reactions. *ACS Catal.* **2016**, *6*, 989−1023.

(4) For selected reviews, see (a) Yu, S.; Ma, S. Allenes in catalytic asymmetric synthesis and natural product syntheses. *Angew. Chem., Int. Ed.* **2012**, *51*, 3074−3112. (b) Liu, L.; Ward, R. M.; Schomaker, J. M. Mechanistic aspects and synthetic applications of radical additions to allenes. *Chem. Rev.* **2019**, *119*, 12422−12490. (c) Jeganmohan, M.; Cheng, C.-H. Transition metal-catalyzed three-component coupling of allenes and the related allylation reactions. *Chem. Commun.* **2008**, 3101−3117.

(5) For a review, see Pulis, A. P.; Yeung, K.; Procter, D. J. Enantioselective copper catalysed, direct functionalisation of allenes via allyl copper intermediates. *Chem. Sci.* **2017**, *8*, 5240−5247.

(6) Meng, F.; Jang, H.; Jung, B.; Hoveyda, A. H. Cu-catalyzed chemoselective preparation of 2-(pinacolato)boron substituted allylcopper complexes and their *in situ* site-, diastereo-, and enantioselective additions to aldehydes and ketones. *Angew. Chem., Int. Ed.* **2013**, *52*, 5046−5051.

(7) (a) Yeung, K.; Ruscoe, R. E.; Rae, J.; Pulis, A. P.; Procter, D. J. Enantioselective generation of adjacent stereocenters in a copper-catalyzed three-component coupling of imines, allenes, and diboranes. *Angew. Chem., Int. Ed.* **2016**, *55*, 11912−11916. (b) Jang, H.; Romiti, F.; Torker, S.; Hoveyda, A. H. Catalytic diastereo- and enantioselective additions of versatile allyl groups to *N*-H ketimines. *Nat. Chem.* **2017**, *9*, 1269−1275. (c) Deng, H.; Meng, Z.; Wang, S.; Zhang, Z.; Zhang, Y.; Shangguan, Y.; Yang, F.; Yuan, D.; Guo, H.; Zhang, C. Enantioselective copper-catalyzed three-component carboboronation of allenes: access to functionalized dibenzo [b,f][1,4]oxazepine derivatives. *Adv. Synth. Catal.* **2019**, *361*, 3582−3587.

(8) Zhang, S.; del Pozo, J.; Romiti, F.; Mu, Y.; Torker, S.; Hoveyda, A. H. Delayed catalyst function enables direct enantioselective conversion of nitriles to NH_2-amines. *Science* **2019**, *364*, 45−51.

(9) Meng, F.; McGrath, K. P.; Hoveyda, A. H. Multifunctional organoboron compounds for scalable natural product synthesis. *Nature* **2014**, *513*, 367−374.

(10) (a) Meng, F.; Li, X.; Torker, S.; Shi, Y.; Shen, X.; Hoveyda, A. H. Catalytic enantioselective 1,6-conjugate additions of propargyl and allyl groups. *Nature* **2016**, *537*, 387−393. (b) Huang, Y.; Torker, S.; Li, X.; del Pozo, J.; Hoveyda, A. H. Racemic vinylallenes in catalytic enantioselective multicomponent processes: rapid generation of complexity through 1,6-conjugate additions. *Angew. Chem., Int. Ed.* **2019**, *58*, 2685−2691.

(11) Han, J.; Zhou, W.; Zhang, P.-C.; Wang, H.; Zhang, R.; Wu, H.-H.; Zhang, J. Design and synthesis of WJ-Phos, and application in Cu-catalyzed enantioselective boroacylation of 1,1-disubstituted allenes. *ACS Catal.* **2019**, *9*, 6890−6895.

(12) Cyanoborylation of other unsaturated hydrocarbons: (a) Yang, Y.; Buchwald, S. L. Copper-catalyzed regioselective ortho C-H cyanation of vinylarenes. *Angew. Chem., Int. Ed.* **2014**, *53*, 8677−8681. (b) Yang, Y.; Liu, P. Mechanism and origins of selectivities in the copper-catalyzed dearomatization-induced ortho C−H cyanation of vinylarenes. *ACS Catal.* **2015**, *5*, 2944−2951. (c) Zhao, W.; Montgomery, J. Functionalization of styrenes by copper-catalyzed borylation/ortho-cyanation and silver-catalyzed annulation processes. *Angew. Chem., Int. Ed.* **2015**, *54*, 12683−12686. (d) Yang, Y. Regio- and stereospecific 1,3-allyl group transfer triggered by a copper-catalyzed borylation/ortho-cyanation cascade. *Angew. Chem., Int. Ed.* **2016**, *55*, 345−349. (e) Jia, T.; He, Q.; Ruscoe, R. E.; Pulis, A. P.; Procter, D. J. Regiodivergent copper catalyzed borocyanation of 1,3-dienes. *Angew. Chem., Int. Ed.* **2018**, *57*, 11305−11309. (f) Wen, L.; Zhang, H.; Wang, J.; Meng, F. Cu-catalyzed regioselective borylcyanation of 1,3-dienes. *Chem. Commun.* **2018**, *54*, 12832−12835. (g) He, S.-J.; Wang, B.; Lu, X.; Gong, T.-J.; Yang, Y.-N.; Wang, X.-X.; Wang, Y.; Xiao, B.; Fu, Y. Copper-catalyzed reagent-controlled regioselective cyanoborylation of vinylarenes. *Org. Lett.* **2018**, *20*, 5208−5212. During the preparation of this manuscript, a copper-catalyzed enantioselective borocyanation of 1-aryl-1,3-butadienes for the synthesis of α-borylmethyl allyl nitriles was reported: (h) Jia, T.; Smith, M. J.; Pulis, A. P.; Perry, G. J. P.; Procter, D. J. Enantioselective and regioselective copper-catalyzed borocyanation of 1-aryl-1,3-butadienes. *ACS Catal.* **2019**, *9*, 6744−6750.

(13) Yamamoto, A.; Ikeda, Y.; Suginome, M. Palladium-catalyzed intramolecular cyanoboration of allenes leading to the regioselective synthesis of β-cyanoallylboranes. *Tetrahedron Lett.* **2009**, *50*, 3168−3170.

(14) (a) Zhao, W.; Montgomery, J. Cascade copper-catalyzed 1,2,3-trifunctionalization of terminal allenes. *J. Am. Chem. Soc.* **2016**, *138*, 9763−9766. (b) Kim, H. K.; Mane, M. V.; Montgomery, J.; Baik, M.-H. The mechanism of copper-catalyzed trifunctionalization of terminal allenes. *Chem. − Eur. J.* **2019**, *25*, 9456−9463.

(15) Zhang, L.; Cheng, J.; Carry, B.; Hou, Z. Catalytic boracarboxylation of alkynes with diborane and carbon dioxide by an *N*-heterocyclic carbene copper catalyst. *J. Am. Chem. Soc.* **2012**, *134*, 14314−14317.

(16) Carry, B.; Zhang, L.; Nishiura, M.; Hou, Z. Synthesis of lithium boracarbonate ion pairs by copper-catalyzed multi-component coupling of carbon dioxide, diboron, and aldehydes. *Angew. Chem., Int. Ed.* **2016**, *55*, 6257−6260.

(17) (a) Li, Z.; Zhang, L.; Nishiura, M.; Luo, G.; Luo, Y.; Hou, Z. CO_2 Activation by Lewis pairs generated under copper catalysis enables difunctionalization of imines. *J. Am. Chem. Soc.* **2020**, *142*, 1966−1974. (b) Li, Z.; Zhang, L.; Nishiura, M.; Hou, Z. Copper-catalyzed umpolung of imines through carbon-to-nitrogen boryl migration. *ACS Catal.* **2019**, *9*, 4388−4393.

(18) Neeve, E. C.; Geier, S. J.; Mkhalid, I. A. I.; Westcott, S. A.; Marder, T. B. Diboron(4) compounds: from structural curiosity to synthetic workhorse. *Chem. Rev.* **2016**, *116*, 9091−9161.

(19) (a) Chaitanya, M.; Anbarasan, P. Recent developments and applications of cyanamides in electrophilic cyanation. *Org. Biomol. Chem.* **2018**, *16*, 7084−7103. (b) Cui, J.; Song, J.; Liu, Q.; Liu, H.; Dong, Y. Transition-metal-catalyzed cyanation by using an electrophilic cyanating agent, *N*-cyano-*N*-phenyl-*p*-toluenesulfonamide (NCTS). *Chem. − Asian J.* **2018**, *13*, 482−495.
(20) (a) Luan, X.; Mariz, R.; Gatti, M.; Costabile, C.; Poater, A.; Cavallo, L.; Linden, A.; Dorta, R. Identification and characterization of a new family of catalytically highly active imidazolin-2-ylidenes. *J. Am. Chem. Soc.* **2008**, *130*, 6848−6858. (b) Luan, X.; Mariz, R.; Robert, C.; Gatti, M.; Blumentritt, S.; Linden, A.; Dorta, R. Matching the chirality of monodentate *N*-heterocyclic carbene ligands: a case study on well-defined palladium complexes for the asymmetric α-arylation of amides. *Org. Lett.* **2008**, *10*, 5569−5572. (c) Luan, X.; Wu, L.; Drinkel, E.; Mariz, R.; Gatti, M.; Dorta, R. Highly chemo- and enantioselective synthesis of 3-allyl-3-aryl oxindoles via the direct palladium-catalyzed α-arylation of amides. *Org. Lett.* **2010**, *12*, 1912−1915. (d) Wu, L.; Falivene, L.; Drinkel, E.; Grant, S.; Linden, A.; Cavallo, L.; Dorta, R. Synthesis of 3-fluoro-3-aryl oxindoles: direct enantioselective α-arylation of amides. *Angew. Chem., Int. Ed.* **2012**, *51*, 2870−2873.
(21) Le, H.; Kyne, R. E.; Brozek, L. A.; Morken, J. P. Catalytic enantioselective allyl-allyl cross-coupling with a borylated allylboronate. *Org. Lett.* **2013**, *15*, 1432−1435.
(22) CCDC 1942085 ((*R*)-**3b**) contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre.
(23) Shade, R. E.; Hyde, A. M.; Olsen, J.-C.; Merlic, C. A. Copper-promoted coupling of vinyl boronates and alcohols: a mild synthesis of allyl vinyl ethers. *J. Am. Chem. Soc.* **2010**, *132*, 1202−1203.
(24) Yeung, K.; Talbot, F. J. T.; Howell, G. P.; Pulis, A. P.; Procter, D. J. Copper-catalyzed borylative multicomponent synthesis of quaternary α-amino esters. *ACS Catal.* **2019**, *9*, 1655−1661.
(25) (a) Laitar, D. S.; Müller, P.; Sadighi, J. P. Efficient homogeneous catalysis in the reduction of CO_2 to CO. *J. Am. Chem. Soc.* **2005**, *127*, 17196−17197. (b) Zhao, H.; Lin, Z.; Marder, T. B. Density functional theory studies on the mechanism of the reduction of CO_2 to CO catalyzed by copper(I) boryl complexes. *J. Am. Chem. Soc.* **2006**, *128*, 15637−15643. (c) Dang, L.; Lin, Z.; Marder, T. B. Boryl ligands and their roles in metal-catalysed borylation reactions. *Chem. Commun.* **2009**, 3987−3995.
(26) Semba, K.; Shinomiya, M.; Fujihara, T.; Terao, J.; Tsuji, Y. Highly selective copper-catalyzed hydroboration of allenes and 1,3-dienes. *Chem. − Eur. J.* **2013**, *19*, 7125−7132.
(27) The simulation of the 3,2-insertion transition state is finally converted to the 1,2-insertion transition state due to its extremely unstable feature. The energy of **TS1**^{c} was estimated by an approximate structure, which is located in the flat transition state region. Please see Figure S1.
(28) The preliminary calculation results revealed that further borocupration of **2a** is kinetically feasible. However, the catalytic cycle would not be accessible in the absence of an additional proton source. This is in line with the experimental observation that trifunctionalization could take place when MeOH was added.[14]

https://dx.doi.org/10.1021/acscatal.0c03018
ACS Catal. 2020, 10, 11685−11692